<u>Exam Review paper</u>

Q1. A force in the negative direction of an x- axis is applied for 27×10^{-3} s to a 0.40 kg ball initially moving at 14.0 m/s in the positive direction of the x-axis. The force varies in magnitude and the impulse has magnitude 32.4 N.s.
(a) What is the ball's speed and direction of travel just after the force is removed?
(b) What is the average magnitude of the force and the direction of the impulse on the ball?
0
Q2. A ice block of mass 10 kg floating in a river is pushed through a displacement **d** given by **d** = (20 m) **i** – (16 m) **j** along a straight embankment by rushing water, which exerts a force **F** = (210 N) **i** – (150 N) **j** on the block.
(a) How much work does the force do on the block during the displacement?
(b) What is the magnitude of the final velocity of the block if it was initially at rest?

Q3. Four point particles each of mass m are kept at the four corners of a square of edge length a. Find the moment of inertia of the system about a line perpendicular to the plane of the square and passing through the centre of the square.

Q4. A wheel rotating at an angular speed of 20 rad/s is brought to rest by a constant torque in 4.0 s. If the moment of inertia of the wheel about the axis of rotation is 0.20 kg-m^2, find (a) The angular deceleration of the wheel during 4.0 s.
(b) The angle rotated in the first two seconds.

Q5. An object of mass 2.4 kg moves up a plane that is inclined at 37.0^0 to the ground. If the initial speed of the object is 3.8 m/s and the coefficient of kinetic friction between the object and the plane is 3/10, how far does the object travel before it stops?

Q6. A force of 30 N is applied to a body of mass 10 kg which was originally at rest. If the object moved 24 m in the same direction as the force, calculate. i) the work done on the body and its final kinetic energy, ii)the final velocity of the body, iii) the rate of work done by the force.

Q7. At t = 0 a grinding wheel has an angular velocity of 24.0 rad/s. It has a constant angular acceleration of 30.0 rad/s^2 until t = 2.0 s. After that there is a constant angular deceleration until it comes to stop. In this period of constant angular deceleration it turns around 432 radians.
(i) Through what total angle did the wheel turn between t = 0 and the time it stopped?
(ii) At what time did it stop?
(iii) What was its angular deceleration as it slowed down?

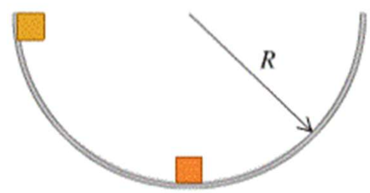
Q8. A stationery object suddenly bursts apart into two pieces. Piece A has mass m_A and moves to the left with speed v_A while piece B has mass m_B moves to the right with speed v_B.
i) Obtain an expression for v_B in terms of m_A, v_A, and m_B.
ii) Find the ratio K_A/K_B where K_A and K_B are the kinetic energies of A and B.

Q9. An object is undergoing simple harmonic motion with period 0.300 s and amplitude 6.00 cm. At $t = 0$, the object is at $x = 6.00$ cm. Calculate the time it takes the object to go from $x = 6.00$ cm to $x = -1.50$ cm?

Q10. A bungee jumper with mass 65.0 kg jumps from a high bridge. After reaching his lowest point, he oscillates up and down, hitting a low point eight times in 43.0 s. He finally comes to rest 25.0 m below the level of the bridge. Estimate (i) the spring stiffness constant and (ii) the unstretched length of the bungee cord assuming SHM.

Q11. A particle executes simple harmonic motion as $x = 3 \sin(\pi t + \pi/3)$ where the amplitude is in metres. What are the amplitude, time period and maximum speed. Find its velocity at $t = 2$s.

Q12. Two identical masses are released from rest in a smooth hemispherical bowl of radius R from the positions shown. You can ignore friction between the masses and the surface of the bowl. If they stick together when they collide, how high above the bottom of the bowl will the masses go after colliding?



Q13. An electric fan is turned off, and its angular velocity decreases uniformly from 500 rev/min to 200 rev/min in 4.00 s. (a) Find the angular acceleration in rev/s2 and the number of revolutions made by the motor in the 4.00-s interval. (b) How many more seconds are required for the fan to come to rest if the angular acceleration remains constant at the value calculated in part (a)?

Q14. At $t = 0$ a grinding wheel has an angular velocity of 24.0 rad/s. It has a constant angular acceleration of 30.0 rad/s^2 until a circuit breaker trips at $t = 2.00$ s. From then on, it turns through 432 rad as it coasts to a stop at constant angular acceleration.
(a) Through what total angle did the wheel turn between $t = 0$ and the time it stopped?
(b) At what time did it stop? (c) What was its angular acceleration as it slowed down?

Q15. A safety device brings the blade of a power mower from an initial angular speed of ω1 to rest in 1.00 revolution. At the same constant angular acceleration, how many revolutions would it take the blade to come to rest from an initial angular speed ω3 that was three times as great, $\omega_3 = 3\omega_1$?

Q16. A child is pushing a merry-go-round. The angle through which the merry-go-round has turned varies with time according $\theta(t) = \gamma t + \beta t^3$ where $\gamma = 0.400$ rad/s and $\beta = 0.0120$ rad/s^3. (a) Calculate the angular velocity of the merry-go-round as a function of time. (b) What is the initial value of the angular velocity? (c) Calculate the instantaneous value of the angular velocity ω at $t = 5.00$ s and the average angular velocity ωav for the time interval $t = 0$ to $t = 5.00$ s. Show that ω_{av} is not equal to the average of the instantaneous angular velocities at $t = 0$ to $t = 5.00$ s and explain why it is not.